

FBISE MATHEMATICS CLASS 9

Chapter 6: Trigonometry and Bearing

COMPREHENSIVE CHAPTER TEST

Prepared by

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Section A - 20 MCQs	20 marks
Section B - 20 Short Questions	80 marks
Section C - Q3 to Q12 (10 Long Questions)	80 marks
Bonus Long Question	8 bonus marks

Regular Total: 180 marks | Bonus: 8 marks

Student Name: _____	Roll No.: _____
Date: _____	Class/Section: _____

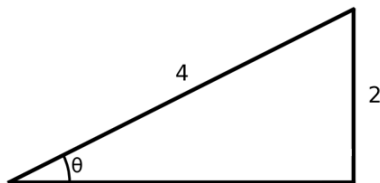
General Instructions

- Attempt all regular questions.
- Each MCQ carries 1 mark; each short question carries 4 marks; each long question carries 8 marks.
- Show all necessary working in Sections B and C.
- Diagrams are not necessarily drawn to scale.
- The bonus long question is optional and carries 8 bonus marks.

SECTION A - Objective

Q1. Choose the correct option. ($20 \times 1 = 20$ marks)

1. From the given right-angled triangle, find θ .



A) 90°	B) 60°	C) 30°	D) 0°
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2. If a plane flies on a bearing of 180° , it is moving towards:

A) North	B) South	C) East	D) West
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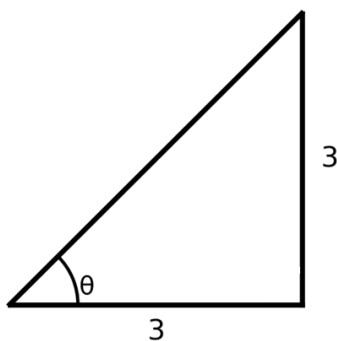
3. If two triangles are equal, then their corresponding sides are:

A) Equal	B) Unequal	C) Unproportional	D) Perpendicular
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4. $1/\sin \theta =$

A) $\tan \theta$	B) $\cos \theta$	C) $\sec \theta$	D) $\operatorname{cosec} \theta$
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5. In the given right-angled triangle, find θ .



A) 45°	B) 60°	C) 90°	D) 120°
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6. A cyclist travels from P to Q on a bearing of 45° . This means the cyclist moves:

A) South-East	B) North-East	C) South-West	D) North-West
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7. $1 + \tan^2 \theta =$

A) $\sin^2 \theta$	B) $\cos^2 \theta$	C) $\sec^2 \theta$	D) $\operatorname{cosec}^2 \theta$
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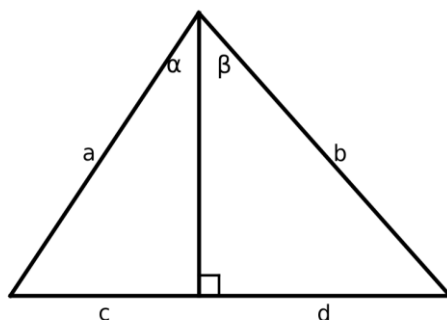
8. What is the radian measure of $15^\circ 50'$?

A) $19\pi/216$	B) $19\pi/36$	C) $19\pi/180$	D) $216\pi/19$
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9. If a navigator gives bearing 0° , in which direction should he travel?

A) North	B) South	C) East	D) West
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10. In the figure, if $\alpha = \beta$, what is the value of b ?

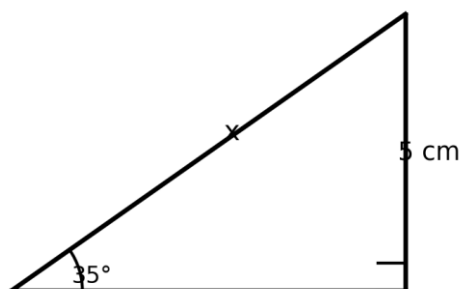


A) cd/a	B) $c/(ad)$	C) ad/c	D) ac/d
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11. What is the value of $-3 - 3\tan^2\theta$ in a single trigonometric function?

A) $3\operatorname{cosec}^2\theta$	B) $-3\sec^2\theta$	C) $3\sec^2\theta$	D) $-3\operatorname{cosec}^2\theta$
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12. What is the value of x in the given figure?



A) 5 cm	B) 10 cm	C) 7.5 cm	D) 15 cm
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13. A ship sails on a three-figure bearing of 075° . In which direction is the ship heading?

A) North	B) East	C) North-East	D) South-East
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14. If the bearing of B from A is 135° , what direction is it?

A) North-West	B) South-East	C) South-West	D) North-East
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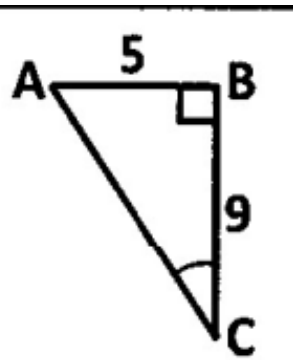
15. Two triangles having equal corresponding angles are called:

A) Similar triangles	B) Right triangles	C) Scalene triangles	D) Isosceles triangles
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16. Which of the following identities is correct?

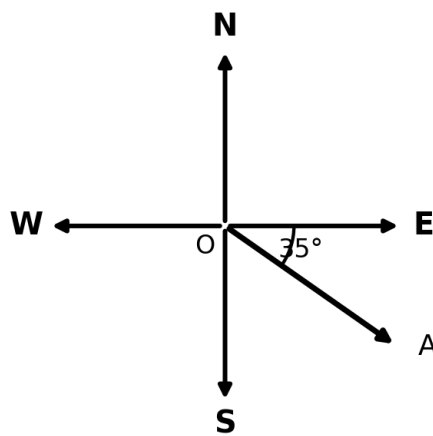
A) $\sin^2\theta + \cos^2\theta = 2$	B) $\sin^2\theta + \cos^2\theta = 1$	C) $\tan^2\theta = 1 + \sec^2\theta$	D) $\cos^2\theta - \sin^2\theta = 0$
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17. What is the measure of angle C in the given figure?



A) 29.1°	B) 60.94°	C) 56.25°	D) 33.75°
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18. The bearing of A from O is:



A) 035°	B) 055°	C) 125°	D) 305°
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19. If an automobile travels on a bearing of 070° from city A to city B, then the bearing from city B to city A is:

A) 070°	B) 110°	C) 250°	D) 290°
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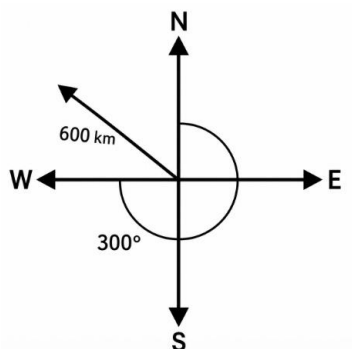
20. A boat sails 4 km north from L to M, then 3 km west from M to P. What is the bearing of P from L, to the nearest degree?

A) 037°	B) 053°	C) 217°	D) 323°
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SECTION B - Short Questions

Q2. Answer all 20 short questions. Each question carries 4 marks. ($20 \times 4 = 80$ marks)

Q2(1). A plane travels 600 km on a bearing of 300° . How far west did the plane travel? [4 marks]



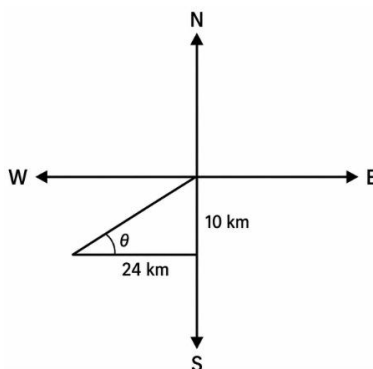
Q2(2). Solve the right-angled triangle $\triangle ABC$, given $\angle C = 90^\circ$, $a = 4$ cm, and $c = 4\sqrt{2}$ cm. [4 marks]

Q2(3). Verify that $\sin^2\theta + \cos^2\theta = 1$. [4 marks]

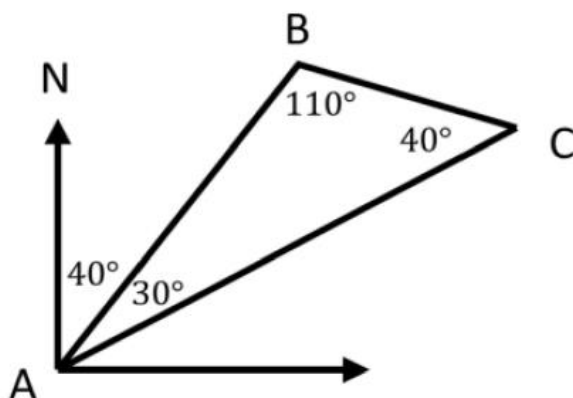
Q2(4). Two boats A and B are 5 km apart, and the bearing of B from A is 250° . Draw a diagram showing their relative positions, then find the bearing of A from B. [4 marks]

Q2(5). Show that $1 + \tan^2\theta = \sec^2\theta$. [4 marks]

Q2(6). A cyclist travels 10 km due south and then 24 km due west. What simple bearing should the cyclist take to return directly to the starting point? [4 marks]



Q2(7). The figure shows the positions of A, B and C. State the bearing of: (i) B from A, (ii) A from B, (iii) B from C, and (iv) C from B. [4 marks]



Q2(8). Prove that $1/(1 + \cos x) + 1/(1 - \cos x) = 2 + 2\cot^2x$. [4 marks]

Q2(9). Write each in three-digit bearing notation: (a) South 55° West, (b) North 15° East, (c) North 110° West, (d) West 35° South. [4 marks]

Q2(10). Prove that $\sin\theta/(1 - \cos\theta) - (1 + \cos\theta)/\sin\theta = 0$. [4 marks]

Q2(11). A person is standing 20 m away from a tree. The angle of elevation from the person's eye level to the top of the tree is 45° . (a) Find the height of the tree. (b) If the person moves 10 m closer, find the new angle of elevation. [4 marks]

Q2(12). A person travels 100 km due East, then 50 km due South. Find the direction and bearing of the final position from the starting point. [4 marks]

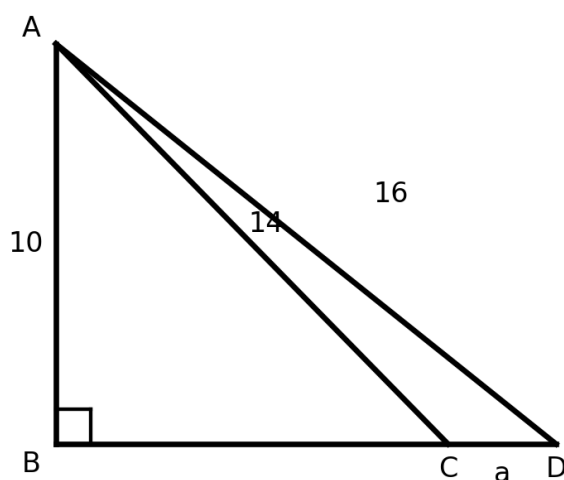
Q2(13). Two trees are 50 ft apart. A lamp post is to be placed 35 ft from each tree. Draw a sketch showing where the lamp post can be placed. How many possible locations are there? [4 marks]

Q2(14). State the three-digit bearing for a ship heading: (a) East, (b) South-East, (c) West, (d) North-West. [4 marks]

Q2(15). Prove that $\sin\theta/(1 - \cos\theta) = \operatorname{cosec}\theta + \cot\theta$. [4 marks]

Q2(16). A 10 m long ladder leans against a vertical wall. Its foot is 6 m from the base of the wall. Find: (a) the angle the ladder makes with the ground, (b) the height at which the ladder touches the wall. [4 marks]

Q2(17). Find the value of a in the following figure. [4 marks]



Q2(18). If the shadow of a building increases by 12 m when the angle of elevation of the sun decreases from 60° to 45° , find the height of the building. [4 marks]

Q2(19). A plane is 119 km west and 100 km south of an airport. At what bearing should the pilot fly directly back to the airport? [4 marks]

Q2(20). Prove that $\sqrt{[(1 - \cos\theta)/(1 + \cos\theta)]} = \sin\theta/(1 + \cos\theta)$. [4 marks]

SECTION C - Long Questions

Answer Q3 to Q12. Each question carries 8 marks. ($10 \times 8 = 80$ marks)

Q3. Two men on opposite sides of a tree are in line with it. The angles of elevation of the top of the tree are 30° and 60° . The height of the tree is 15 m. Find the distance between the men. [8 marks]

Q4. From the eye level of a 6-ft-tall person standing on the ground, the angle of elevation of the top of a tower is 30° . (a) What type of triangle is formed? (b) Which trigonometric ratio would be used to find the height of the tower? (c) Find the height of the tower from its base. [8 marks]

Q5. The tide height is modeled by $H = H_0 + A \sin(2\pi t/T)$, where H_0 is average tide height, A is amplitude, t is time in hours, and T is the period. If $H_0 = 2$ m, $A = 1$ m, $T = 24$ h, find H at $t = 0, 6$ and 18 hours. [8 marks]

Q6. A person sees the top of a tower at an angle of elevation of 30° . The horizontal distance to the tower is 50 m. (a) Find the tower height to two decimal places. (b) A bird directly above the top is seen at 60° . Find the vertical distance of the bird from the ground. [8 marks]

Q7. A ship sails from port A to port B on a bearing of 040° for 35 km. It then sails from B to C on a bearing of 130° for 50 km. Find: (a) the distance AC, (b) the bearing of C from A. [8 marks]

Q8. An aeroplane at an altitude of 900 m observes two ships sailing towards it in the same direction. Their angles of depression are 30° and 60° . Find the distance between the ships. [8 marks]

Q9. A boat moving away from a lighthouse 206.6 m high takes 120 seconds for the angle of elevation of the top to decrease from 60° to 45° . Find the speed of the boat. [8 marks]

Q10. From point A, 200 m from a tower, the angle of elevation is α . Point B is 80 m nearer the tower, and its angle of elevation is β . If $\tan \alpha = 2/5$, find (a) the tower height and (b) β . [8 marks]

Q11. From the top of a 12 m tree, the angles of elevation and depression of the top and bottom of a building are 30° and 45° , respectively. Find (a) the height of the building and (b) the distance between the tree and building. [8 marks]

Q12. Two pillars of equal height stand on opposite sides of a 120 m wide roadway. From a point between them, the angles of elevation of their tops are 60° and 30° . Find the height of each pillar and the position of the point. [8 marks]

BONUS LONG QUESTION

Bonus. From the top of a tower 120 m high, the angles of depression of two boats on the same side at water level are 60° and 45° . Find the distance between the boats. [8 marks]

— End of Test —